

Operational guide for deployment, security, storage, audio processing, client and operations.

MULTITRACK DIGITAL AUDIO WORKSTATION



Digital Audio Workstation with



Windows



macOS



Linux

Operational guide for deployment, security, storage, backups, desktop/mobile clients, audio processing, MTA format, observability and releases.

Contents

1. Architecture
2. Deployment
3. Configuration
4. Storage and backups
5. Security and users
6. Audio pipeline
7. Desktop and mobile clients
8. MTA format
9. Operations and observability
10. Upgrade and rollback
11. CI/CD and releases
12. Administrator troubleshooting

1. Architecture

MTA Audio Editor uses a FastAPI backend and a web frontend shared by browsers, desktop clients and mobile apps. Rendering, waveform processing, Demucs and MTA generation run on the backend.

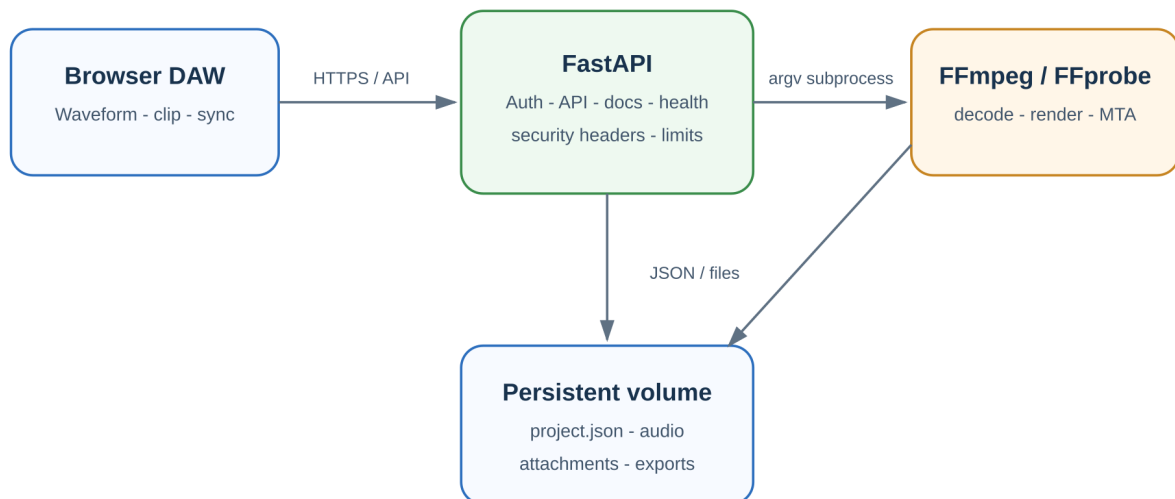


Figure 1 - Clients, API, processing and persistent storage.

Component	Platforms
Server	Linux, Docker, Kubernetes
Desktop	Windows, macOS Intel, macOS Apple Silicon
Mobile	Android, iOS/iPadOS
Browser	Modern browsers with Web Audio API

2. Deployment

Docker Compose should use a persistent volume for `/data/projects`. Kubernetes uses the supplied PVC and an optional Ingress. Application and Ingress upload limits must remain consistent.

```
docker compose up -d
# Kubernetes
kubectl apply -k k8s/overlays/nginx
```

TLS: production deployments should terminate HTTPS at a trusted reverse proxy or Ingress. Configure a stable public URL when email links or OAuth callbacks are enabled.

3. Configuration

Primary settings are supplied through environment variables and Secrets. Administrators should manage bootstrap credentials, upload size, SMTP, OAuth, session security and storage independently.

Area	Examples	Notes
Bootstrap	MTA_ADMIN_USERNAME, MTA_ADMIN_PASSWORD, MTA_ADMIN_EMAIL	Create the initial administrator
Upload	MTA_MAX_UPLOAD_MB	Keep proxy and application limits aligned
Public URL	MTA_PUBLIC_URL	Recommended behind proxies for stable callbacks/links
SMTP	host, port, TLS, credentials	Store credentials in Secrets

4. Storage and backups

The persistent root contains authentication/configuration data and project directories. Each project can contain its model, audio, attachments, originals, caches and analysis outputs. A full recovery backup must include the complete persistent root.

Recommended strategy: use consistent volume snapshots, keep multiple generations and test restores regularly. On Kubernetes verify that the StorageClass/CSI implementation supports the UID/GID and fsGroup used by the pod.

5. Security and users

Server deployments support local accounts, email verification, administrator approval, optional TOTP and optional OAuth providers. Social providers are disabled by default. OAuth requests are restricted to HTTPS and approved provider hosts.

Native single-user desktop mode bypasses the account database and login screens while retaining the internal authorization context required by local API paths. Secrets and signing material must never be baked into container images.

6. Audio pipeline



Editing non distruttivo: le sorgenti restano intatte fino al rendering finale.

Figure 2 - Non-destructive editing and rendering flow.

FFmpeg and Demucs operations can be CPU/GPU intensive. Track previews are cached with signatures derived from source and relevant processing state. Long-running jobs expose progress and final artifacts through the API.

6.1 Dynamic preview and Render

Dynamic preview buffers track sources on the client and keeps them aligned to a reference clock. Render produces a server-side master. With RealTime meters enabled, track monitor branches feed analysers without becoming audible.

7. Desktop and mobile clients

Desktop clients include a local backend and native file pickers. Mobile apps connect to a remote backend.

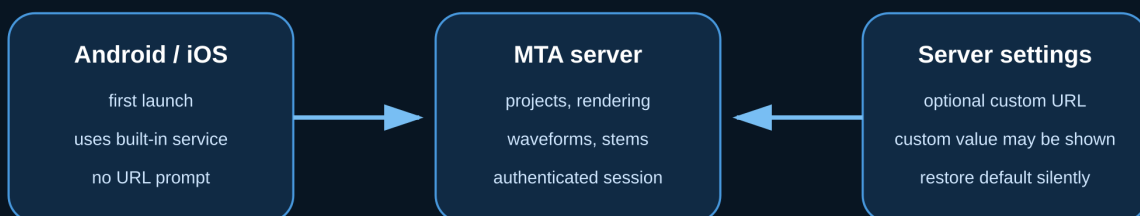


Figure 3 - Mobile connection and custom server settings.

Android and iOS automatically use the preconfigured service without prompting during installation or first launch. The built-in default address remains internal and is never displayed. A custom URL can be configured later and may be shown because it is explicitly user-selected.

8. MTA format

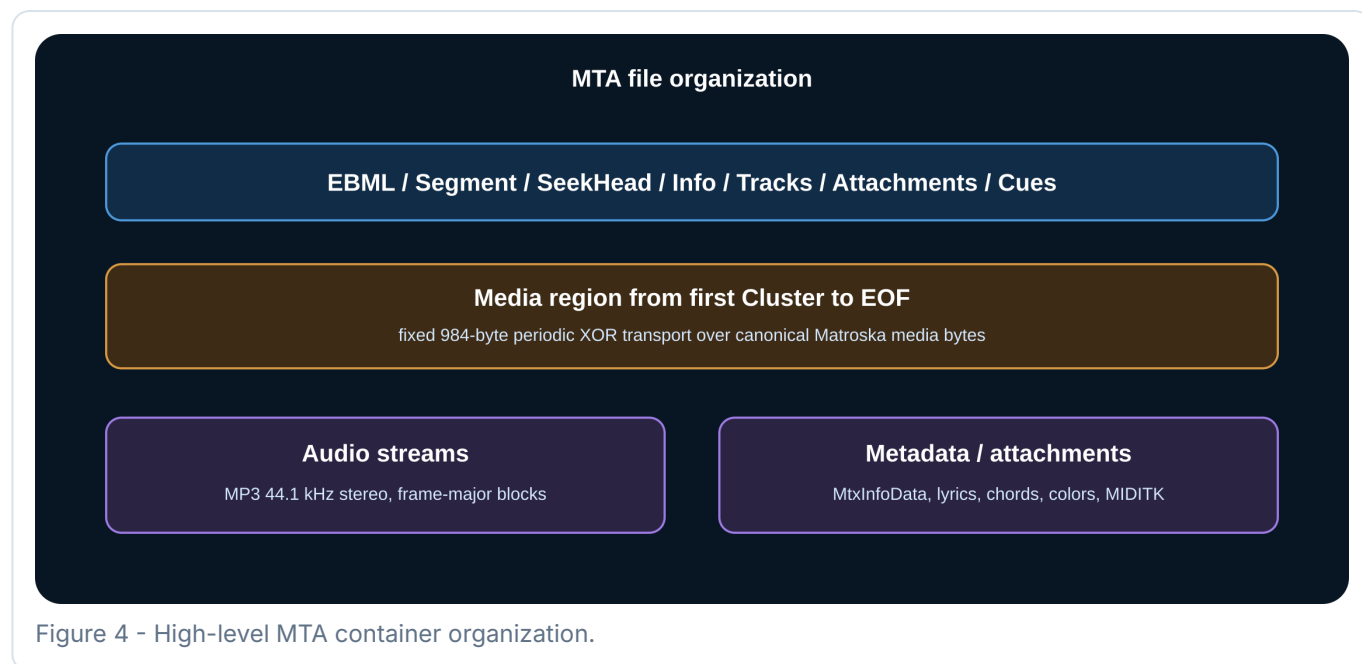
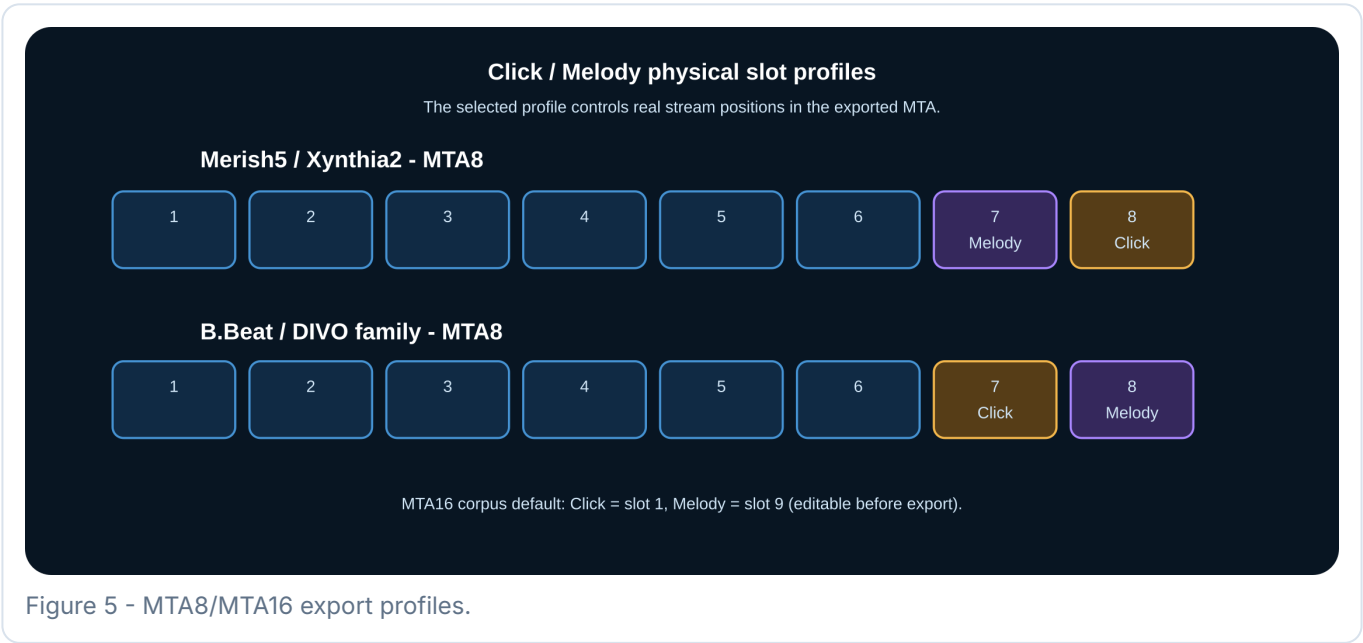


Figure 4 - High-level MTA container organization.

MTA uses EBML/Matroska structures, MP3 audio streams and metadata attachments. The media region starting at the first Cluster uses the documented periodic XOR transport. The complete bilingual specification is stored in docs/MTA_FORMAT_FINAL_SPEC_IT.md and docs/MTA_FORMAT_FINAL_SPEC_EN.md.

8.1 Click/Melody profiles



MTA8 profiles preserve device-specific physical positions. The corpus-based MTA16 default uses Click 1 and Melody 9 and remains editable. Silent intermediate slots are generated when required to preserve the real stream number.

9. Operations and observability

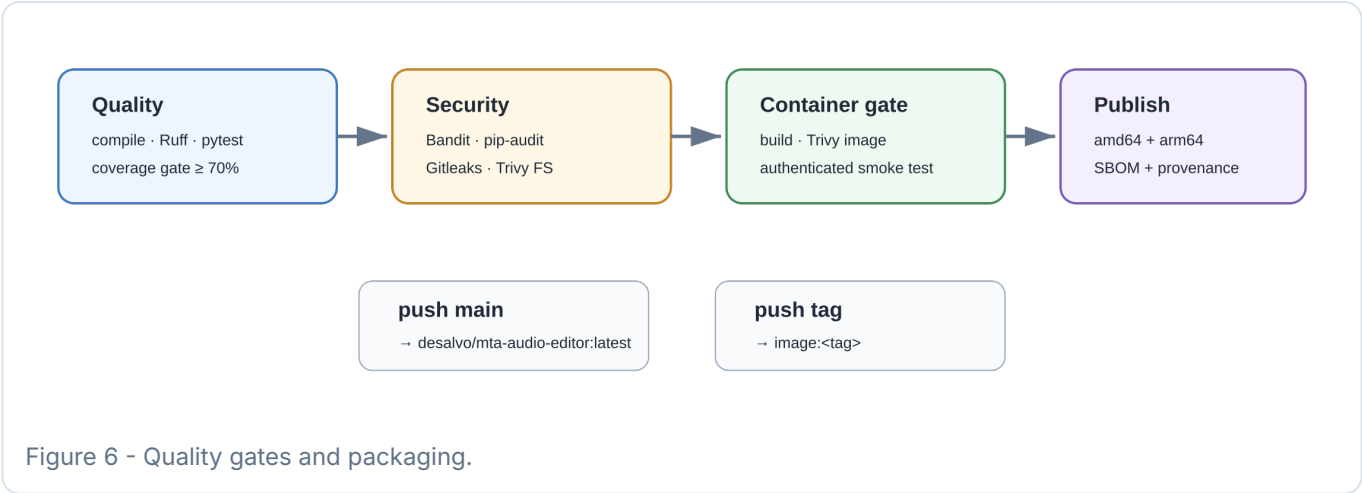
Monitor health endpoints, persistent capacity, render duration, long jobs, application logs, FFmpeg/Demucs failures and service availability. Kubernetes probes should remain lightweight and independent from audio jobs.

Signal	Check
Health	/api/health
Storage	Free space and inodes
Jobs	Duration, state and server-side errors
Authentication	Login/OAuth/SMTP failures
Media	FFmpeg, Demucs, waveform/export logs

10. Upgrade and rollback

Before upgrading, create a backup, review release notes and production-gate results, update images/manifests and validate opening existing projects. For rollback, restore the previous application version and, when data compatibility requires it, the matching persistent-volume snapshot.

11. CI/CD and releases



The production gate includes linting, tests, security checks and packaging validation. Native/mobile artifacts use dedicated build steps. Signing credentials belong in CI Secrets and must not be committed.

12. Administrator troubleshooting

Symptom	Check
413 / upload rejected	Align application and proxy/Ingress upload limits
Stem job is very slow	CPU/GPU, RAM, temporary space and Demucs model
Login/email failure	SMTP, public URL, authentication logs and user status
Pod cannot write to PVC	Storage permissions, UID/GID and fsGroup
Mobile custom server unavailable	Custom URL, TLS, DNS and reachability from the device
MTA rejected by hardware	Slot profile, target capacity, firmware and target-device validation